Deployment Profiles: Evaluated Configurations from Part 2

In Part 2, we evaluated specific configurations of the Cognitive Integrity Framework to understand how different tuning parameters affected security and performance outcomes. The following profiles are derived directly from the **Parameter Sensitivity Analysis** (Part 2, Section 5.3) and **Architecture-Specific Results** (Part 2, Section 5.1).

Profile A: The "Internal Tool" Baseline (Low Latency)

This profile corresponds to the "High Usability" configuration tested in the sensitivity analysis ($\delta = 0.95$). It is designed for low-risk, human-in-the-loop environments.

Configuration Parameters:

- **Trust Decay (δ)**: `0.95`. Maintained $>$50% trust retention even after 13 delegation hops.
- **Firewall Sensitivity**: Relaxed ($\tau = 0.9$).
- **Consensus**: Simple Majority.

Observed Performance (Sensitivity Table 5.2):

- **Latency Overhead**: Minimal ($\sim$15% baseline).
- **Detection Rate**: **87%** (vs 94% optimal).

Architecture-Specific Observations

Beyond the parameter profiles, Part 2's architecture adapters revealed specific interactions between the defense framework and the underlying agent topology.

LangGraph (State Machines)

Observation: LangGraph architectures achieved the highest overall detection rates (98%) in our tests (Part 2, Table 5.6).
Mechanism: The explicit definition of state transitions allowed for rigorous **Invariant Checking**. Invalid state transitions were detected deterministically by the framework.

CrewAI (Role-Based)

Observation: CrewAI architectures performed best against "Trust Exploitation" attacks (94% detection) (Part 2, Table 5.4).
Mechanism: The framework's role definitions acted as implicit **Identity Tripwires**. When an agent attempted to act outside its defined role, the behavior was flagged as a role violation.

Minimal Viable Implementation

We also evaluated a "Minimal Viable Implementation" (MVI) to determine the baseline efficacy of the framework's core components.

The Setup:

1. **Trust Decay**: $\delta = 0.80$ (The optimal balance point).
2. **Cognitive Firewall**: Ingress only.
3. **Tripwires**: One per agent.

Result: Even this minimal setup shifted the success rate against low-effort attacks from 100% (Baseline) to <5%, providing a critical first line of defense.